INTERIM FEDERAL SPECIFICATION

PLOTTER-PROTRACTOR, AIR NAVIGATION

This Interim Federal Specification was developed by the General Services Administration, Federal Supply Service, Washington, D. C., 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers air navigation plotters for use in plotting courses in air navigation and measuring distances in statute miles on charts, regional, and sectional maps.

1.2 Classification.

1.2.1 Types. The plotters shall be furnished in the following types, as specified in the invitation for bids (see 6.1).

Type I. Plotter with offset protractor (see figure 1.)
Type II. Plotter with inset protractor (see figure 2.)

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Box, Fiberboard
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material)
PPP-B-676 - Boxes, Set-UP, Paperboard
L-V-351 - Vinyl Chloride Polymer and Copolymer Rigid Sheets

Federal Standards:
FED.STD. No. 102 - Preservation, Packaging, and Packing Levels
FED.STD. No. 123 - Marking for Domestic Shipment (Civilian Agencies)

(FOR INSTRUCTIONS ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS, SEE LAST PAGE)

Military Specifications:
MIL-P-116 - Preservation, Methods of

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. There are no other publications applicable to this specification.

3. REQUIREMENTS

3.1 Material. Clear, transparent vinyl polymer plastic sheet in accordance with L-V-351, Class I, shall be used in the manufacturer of type I and type II plotters.

3.2 Design

3.2.1 Configuration of type I plotters shall be similar to that shown on figure 1.

3.2.2 Configuration of type II plotters shall be similar to that shown on figure 2.

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3.3 Statute mile scales. Statute mile scales for type I and type II plotters shall be in the following ratios:

Outer scales	1:1,000,000	scales	1 mile - .06336 inches
			100 miles - 6.336 inches
			200 miles - 12.672 inches
Inner scales	1:500,000	scales	1 mile - .12672 inches
			50 miles - 6.336 inches
			100 miles - 12.672 inches

3.4 Type I plotter, scales and protractor

3.4.1 Top outer scale. The top outer scale for type I plotters shall be calibrated in statute mile ratio (see 3.3) from zero to 100 in both directions beginning at the vertical center line of the scale. The calibrations shall be marked in increments of 10.

3.4.2 Bottom outer scale. The bottom outer scale for type I plotters shall be calibrated in statute mile ratio (see 3.3) from zero to 200 reading from left to right facing the plotter. The calibrations shall be marked in increments of 10.

3.4.3 Top inner scale. The top inner scale for type I plotters shall be calibrated in statute mile ratio (see 3.3) from zero to 50 in both directions beginning at the vertical center line of the scale. The calibrations shall be marked in increments of 5.

3.4.4 Bottom inner scale. The bottom inner scale for type I plotters shall be calibrated in statute mile ratio (see 3.3) from zero to 100 reading from left to right facing the plotter. The calibrations shall be marked in increments of 5.

3.4.5 Inch scale.

3.4.5.1 Inch scale. The inch scale shall be of standard measure. It shall be located in the center of the scale. It shall be graduated in eights of an inch reading from left to right facing the plotter, from zero to 12 inches. The inch marks shall be designated by numerals beginning with zero.

3.4.6 Vertical End Scale. The vertical end scale shall be calibrated in statute mile ratio (see 3.3 outer scale). The left end, facing the plotter, shall be calibrated from 1 to 23 and the right end, facing the plotter, shall be calibrated from 23 to 1. The calibrations shall be marked in increments of 3.

3.4.7 Protractor

3.4.7.1 Protractor. The protractor for the type I plotter shall be centrally offset from the top outer scale. The 90 degree marking on the outer semicircle of the protractor, the 270 degree marking of the inner semicircle of the protractor, the zero calibration of the top statute mile scale, the graduation marking for 6 inches, the 50 calibration marking of the bottom inner statutes mile scale and the 100 calibration marking on the bottom statute mile scale, shall all coincide in a vertical center line through the scale and protractor. The outer semicircle of the protractor, reading from right to left facing the plotter, shall be graduated from zero to 180 degrees. The graduations shall be marked every 10 degrees beginning with the 10 degree mark. The inner semicircle of the protractor, reading from left to right facing the plotter, shall be graduated from 360 degrees to 180 degrees. The graduations shall be marked every ten degrees beginning with the 350 degree mark. An arrow pointing to the right shall be placed below the outer semicircle of the protractor centered below the 90 degree graduation marking. An arrow pointing to the left shall be placed below the inner semicircle of the protractor centered below the 270 degree graduation marking. When specified in (6.1), an auxiliary protractor scale shall be located so that it extends 30 degrees on each side of the center line through the 90 degree and 270 degree readings of the main protractor scales. The auxiliary protractor scales shall be subdivided in the same manner as the main protractor scales. The divisions shall be marked at each 10 degree both above and below the scale. The top scale shall be marked reading from left to right facing the plotter 210, 200, 190, 180, 170, 160 and 150 degrees. The lower scale reading from left to right shall be marked 30, 20, 10, 0, 350, 340 and 330 degrees. The arc of the auxiliary protractor scale shall be concentric with the arc of the main protractor scale. This arc shall be 1/2 inch from the inner arc of the main protractor. The numerals on the auxiliary scale shall be the same size as the numerals on the main protractor.

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3.5 Type II plotters; scales and protractor

3.5.1 Top outer scale. The top outer scale for the type II plotters shall be calibrated in statute mile ratio (see 3.3) from zero to 200, reading from left to right facing the plotter. The calibrations shall be marked in increments of 10 to the point of intersect with the inset protractor, and from the right side of the protractor to the end of the plotter. The 90, 100 and 110 marks shall be omitted.

3.5.2 Bottom outer scale. The bottom outer scale for the type II plotters shall be calibrated in statute mile ratio (see 3.3) from zero to 200, reading from right to left facing the plotter. The calibrations shall be marked in increments of 10 to the point of intersect with the inset protractor, and from the left side of the protractor to the end of the plotter. The 70 and 130 marks shall be omitted.

3.5.3 Top inner scale. The top inner scale for type II plotters shall be calibrated in statute mile ratio (see 3.3) from zero to 100, reading from left to right facing the plotter. The calibrations shall be marked in increments of 5 to the point of intersect with the inset protractor, and from the right side of the protractor to the end of the plotter. The 40, 45, 50, 55 and 60 marks shall be omitted.

3.5.4 Bottom inner scale. The bottom inner scale for the type II plotters shall be calibrated in statute mile ratio (see 3.3) from zero to 100, reading from right to left facing the plotter. The calibrations shall be marked in increments of 5 to the point of intersect with the inset protractor, and to the left of the protractor to the end of the plotter. The 35 and 65 marks shall be omitted.

3.5.5 Inch scale.

3.5.5.1 Inch scale. The inch scale shall be of standard measure. It shall be located in the center of the scale. It shall be graduated in eights of an inch with each inch graduation mark designated by numerals beginning with zero, reading from left to right facing the plotter.

3.5.6 Protractor.

3.5.6.1 Protractor. The protractor for the type II plotter shall be centrally inset between the top outer and the bottom outer statute mile scales, intersecting with the 70 and 130 calibration marks of the bottom outer statute mile scale. The 100 calibration mark of the bottom outer scale, the 6 inch mark of the inch scale, and the 90 and 100 degree marks of the protractor shall coincide on a center line through the plotter. The outer semicircle of the protractor shall be graduated from 360 degrees to 180 degrees, reading from left to right facing the plotter. The graduations shall be marked each 10 degrees beginning with 350 degrees, left to right. The inner semicircle of the protractor shall be graduated from 0 to 180 degrees, reading from right to left facing the plotter. The graduations shall be marked each 10 degrees beginning with the 10 degree graduation, right to left. Two arrows pointing to the left, when facing the plotter, shall be placed on the outside of the outer semicircle of the protractor one above the 220 and one above the 320 degree mark. An arrow pointing to the right, when facing the plotter, shall be placed below the 90 degree graduation mark of the inner semicircle of the protractor.

3.6 Type I plotter holes.

3.6.1 Holes. Type I plotters shall have a pivot hole .050 inches in diameter placed on the top outer statute miles scale in the exact center of the plotter length. Two additional holes shall be placed, one at each end of the protractor inner semicircle, and the top outer statute mile scale, and shall be centered at the two intersections of the protractor inner semicircle and the top outer statute mile scale. Another hole shall be placed at the zero mark of the inch scale and its horizontal line, centering at this intersection. The above holes are positioned facing the plotter.

3.7 Type II plotter holes.

3.7.1 Holes. Type II plotters shall have a pivot hole placed on the bottom outer statute mile scale in the exact center of the plotter length. Two additional holes shall be placed, one at each end of the protractor inner semicircle, and the bottom outer statute mile scale, and shall be centered at the intersections of the protractor inner semicircles and the bottom outer statute mile scale. Two additional holes shall be placed, one at each end of the bottom outer statute mile scale, centered above the fifth calibration on the right end facing the plotter, and the 195 calibration on the left end facing the plotter. One hole shall be placed at the intersection of the zero graduation mark and the horizontal line of the inch scale.

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3.8 Type I plotter calibrations and graduations.

3.8.1 Calibrations and graduations; letters, numbers. All calibrations, graduations, letters and numbers shall be uniform and of standard commercial quality. Raised calibrations, graduations, letters, and numbers are not acceptable. The calibration marks for the statute mile ratio scales shall be of two lengths. The shortest length denoting each mile and the longest denoting increments of 5 and 10 miles as applicable. The graduations on the protractor shall be three lengths; the shortest length denoting 1 degree; the medium length denoting each 5 degrees and the longest length being a radial line extending to the center hole, located in the center of the protractor on the horizontal statute mile scale, in the center of the plotter length. The radial lines shall be spaced each 10 degrees. The outer two edges of the plotter shall be beveled approximately at a 45 degree angle. The following ratios shall be printed or otherwise permanently marked upon the plotter below the inner statute mile scale: STATUTE MILES - OUTER and VERTICAL SCALES 1:1,000,000 - INNER SCALES 1:500,000. The calibrations, graduations, letters, numbers, and ratios shall be printed in jet black ink on the underside of the plastic sheet specified in 3.1, and a vinyl overlay sheet of not less than .005 inch thick shall be laminated over the printing to the underside in a Hydraulic Platen Press under the proper temperature and pressure to make a solid homogeneous sheet.

3.9 Type II plotter calibrations and graduations.

3.9.1 Calibrations and graduations; letters, numbers. All calibrations and graduations, letters, and numbers shall be uniform and of standard commercial quality. Raised calibrations, graduations, letters, and numbers are not acceptable. The calibration marks for the statute mile ratio scales shall be of two lengths. The shorter length shall denote each mile and the longer length shall denote increments of 5 and 10 miles as applicable. The graduations on the protractor shall be three lengths; the shortest length denoting 1 degree, the medium length denoting each 5 degrees and the longest length being a radial line extending from the edge of the plotter to the hole located in the center of the protractor on the bottom statute mile ratio scale at the 100 calibration mark. The radial lines shall be spaced each 10 degrees. The following ratios shall be printed or otherwise permanently marked upon the plotter below the inch scale; OUTER SCALES - STATUTE MILES 1:1,000,000 - INNER SCALES STATUTE MILES 1:500,000. The first ratio shall be placed to the left of the protractor and the second ratio to the right of the protractor as illustrated in figure 2. The calibrations, graduations, letters, numbers, and ratios shall be in red ink unless otherwise specified. (see 6.1).

3.10 Dimensions. Type I plotters shall be approximately 13-1/2 inches long, 1-1/2 inches wide across scales and approximately .045 inches thick. The type II plotters shall be approximately 12-7/8 inches long, 2-3/8 inches wide and approximately .040 inches thick.

3.11 Workmanship. The surface and edges of the finished plotters shall be true, smooth, uniform and polished. They shall be free from defects which affect the appearance or which might impair the serviceability or use of the plotters.

3.12 Identification markings. Each plotter shall have the manufacturer's name or trade mark permanently printed or otherwise permanently marked upon the plotter.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection of plotters.
(b) Inspection of preparation for delivery.

4.3 Quality conformance inspection.

4.3.1 Sampling.

4.3.1.1 For examination. Sampling for examination shall be in accordance with MIL-STD-105.

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4.3.2 Examination. Samples selected in accordance with 4.3.1.1 shall be examined as specified in 4.4.1. AQL shall be 4.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The plotters shall be examined for the following defects:

(a) Type not as specified.
(b) Material not as specified.
(c) Calibrations, graduations, letters, numbers, and ratios not as specified.
(d) Dimensions not as specified.
(e) Construction not as specified.
(f) Workmanship not as specified.

4.4.2 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing and marking requirements as required by section 5 of this specification are complied with. Defects shall be scored in accordance with table 1. For examination of contents, the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 6.0 defects per hundred units.

TABLE 1 - Classification of Preparation for Delivery Defects

Examine	Defects
Markings (exterior and interior containers)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as; incomplete closure of container, loose strapping.
	Bulging or distortion of containers.
Packaging and Packing	Packages improperly sealed. Shipping containers not waterproofed. Contents more or less than specified. Shipping containers not tape reinforced.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and applications of levels of packaging and packing shall be in accordance with Fed. Std. No. 102).

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.1).

5.1.1 Unit packaging.

5.1.1.1 Level A. Each type plotter-protractor shall be individually packaged in accordance with MIL-P-116, method IC-1.

5.1.1.2 Level B. Each plotter-protractor shall be placed in an end opening center seam kraft envelope. Six plotter-protractors of one type and description shall be packaged in a box conforming to PPP-B-566, style II, type A with pye locks, class a; or to PPP-B-676, type, class and style optional; or to PPP-B-665, style optional. Each box shall be securely closed to prevent accidental opening.

5.1.1.3 Level C. The plotter-protractors shall be packaged in accordance with the suppliers commercial practice.

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5.1.2 Intermediate packaging. Unless otherwise specified (see 6.1) six packages of plotter-protractors (or 36 each), of one type and description shall be intermediately packaged in a box conforming to PPP-B-566, style and type optional; or to PPP-B-676, type, class and style optional; or to PPP-B-665, style optional. Each box shall be securely closed to prevent accidental opening. When specified (see 6.1) each box shall be waterproofed in accordance with MIL-P-116, method IC-2.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Plotter-protractors, of one type and description, shall be packed in boxes conforming to PPP-B-636, class 2, V3c, style RSC. The gross weight of the box shall not exceed the weight limitations of the box specification. Each box shall be closed, waterproof sealed with tape and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.2 Level B. Eight intermediate packages of plotter-protractors (288 plotter-protractors) shall be packed in a box conforming to PPP-B-636, type I, class I, style RSC. Sealing and closure shall be in accordance with the appendix to the box specification. When quantities less than specified are required (see 6.1) the scales shall be packed in a box conforming to that specified herein.

5.2.3 Level C. The plotter-protractors shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, symbol, and date of this specification.
(b) Type required (1.2.1).
(c) Color of filled calibrations, graduations, letters, numbers, and ratios if other than red (3.9.1).
(d) Auxiliary protractor required (3.4.7.1).
(e) Level of preservation and packaging and level of packing required (5.2.1, 5.2.2, and 5.2.3).

6.2 Figures 1 and 2 show two styles of air navigation and distance plotters which have been found acceptable. However, the figures are included for illustration only and are not intended to preclude the furnishing of other designs which conform to this specification.

6.3 Federal specifications do not include all types, classes, sizes etc., of the commodities indicated by the titles of the specifications or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Drawing Instrements, not otherwise indexed by name.

Carload minimum weight 12,000 pounds, subject to Rule 34,
Uniform Freight Classification.

Motor:

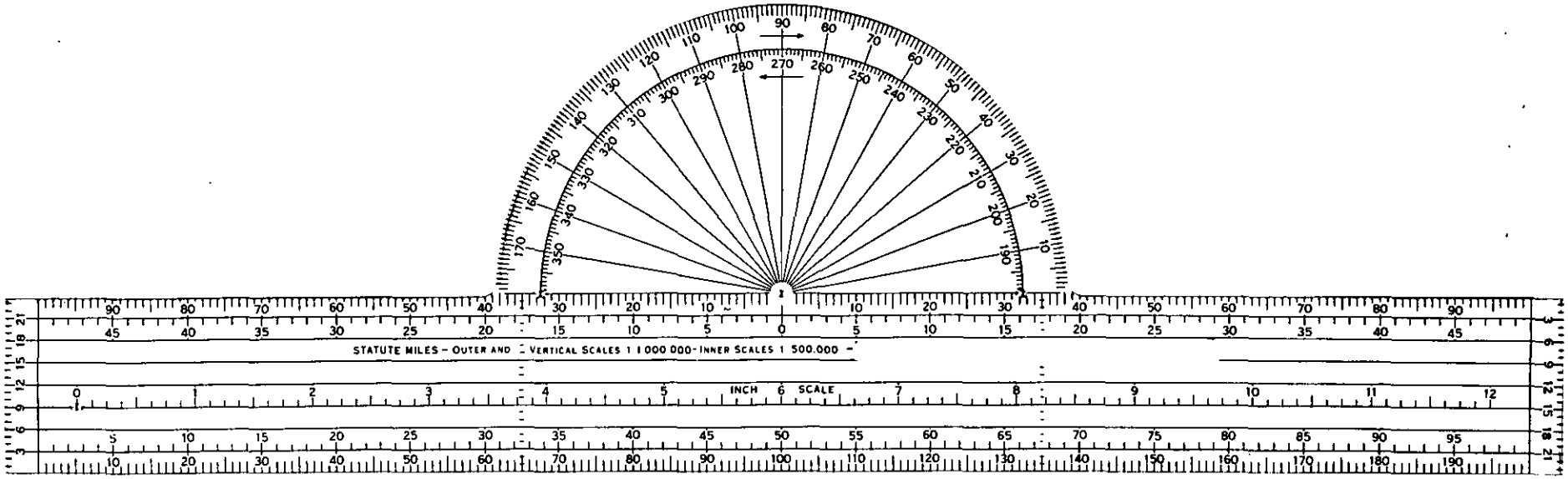
Drawing Instruments, not otherwise indexed.

Truckload minimum weight 12,000 pounds, subject to Rule 115,
National Motor Freight Classification.

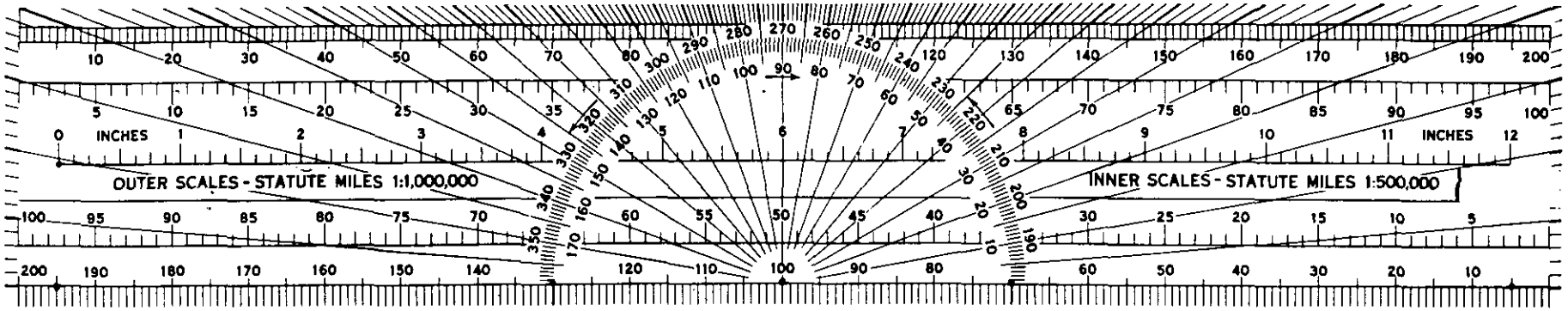
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TYPE I, FIGURE 1



TYPE II, FIGURE 2

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ADDENDUM

INSTRUCTIONS ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

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NOTICE

When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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